VICTORY 2027

Q. 3 marks

Mineral Resources

LECTURE-1 Geography

Sameer Sir



Topics To Be Covered

1. Iron ore, Manganese
2.
3.
4.

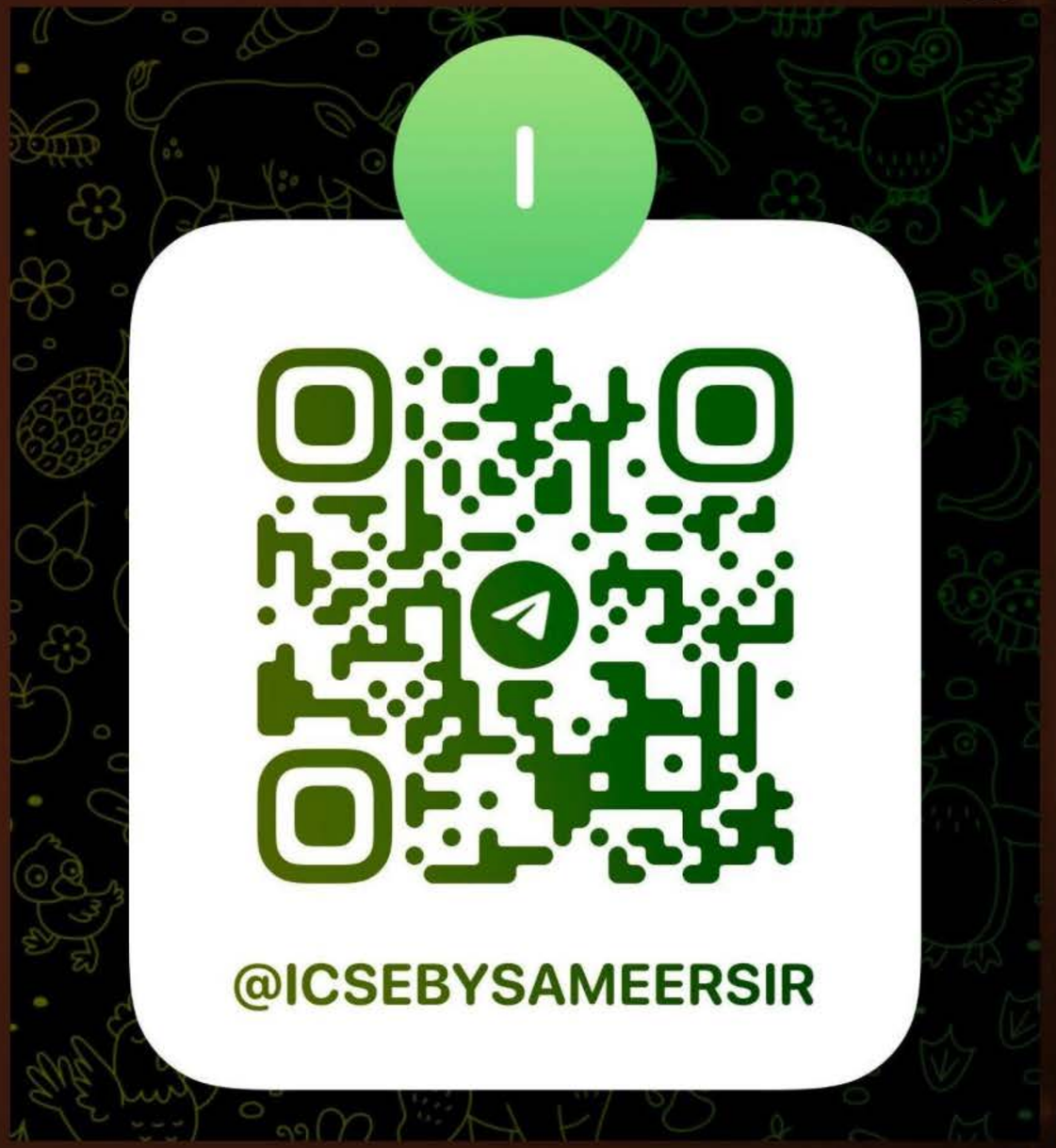
Mineral and Energy Resources

- *Iron ore, Manganese, Copper, Bauxite – uses and their distribution. (two leading states and their prominent mining areas)*
- *Conventional Sources: Coal, Petroleum, Natural gas (distribution, advantages and disadvantages).*
- *Hydel power - Bhakra Nangal Dam and Hirakud Projects and their benefits.*
- *Non-conventional Sources: Solar, wind, tidal, geo-thermal, nuclear and bio-gas (important areas and advantages and disadvantages).*



Telegram

https://t.me/icsebysameersir



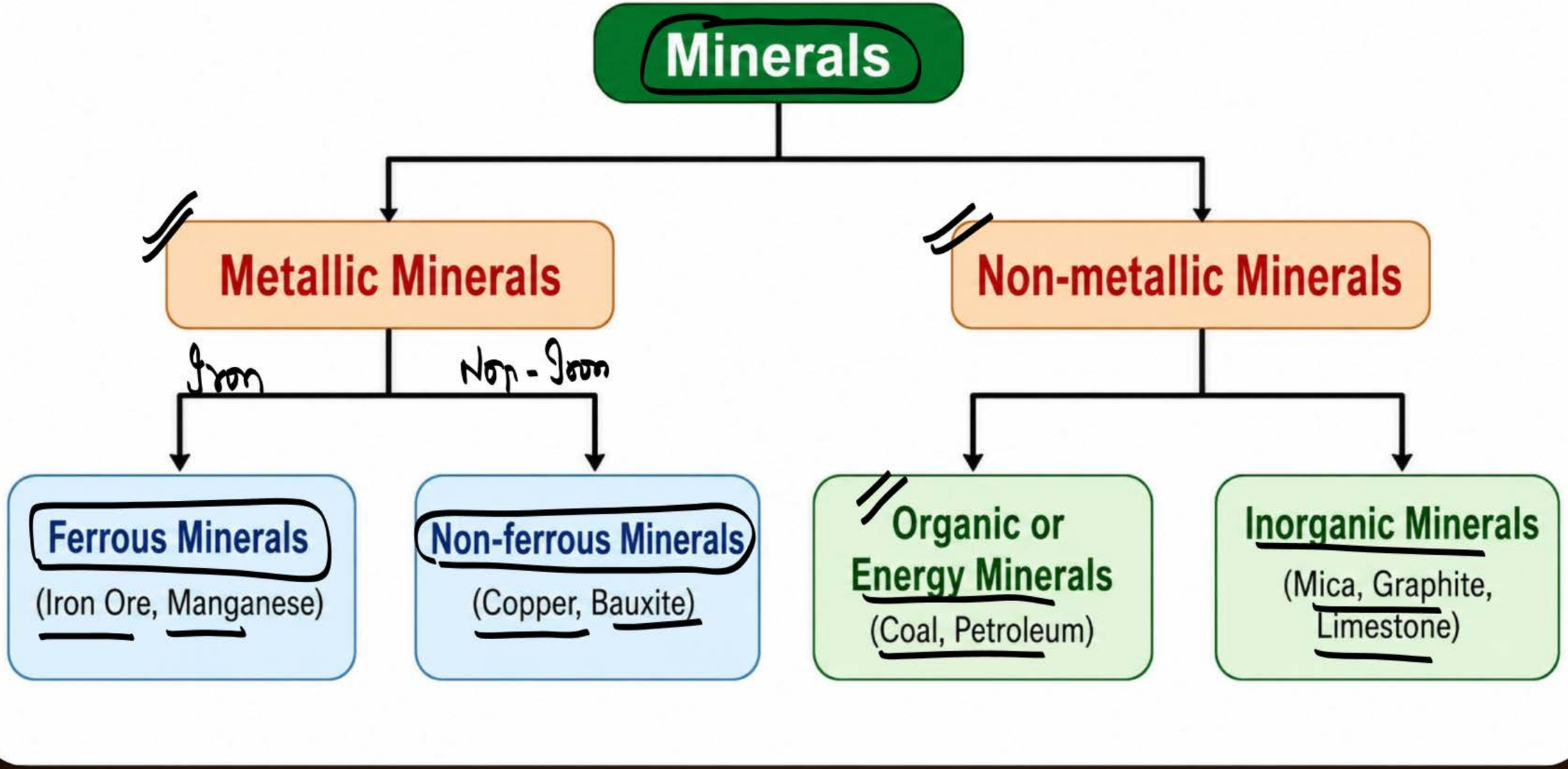


Mineral Resources

Iron - Fe

Manganese →

- Minerals are natural substances of organic or inorganic origin with definite chemical and physical properties.
- Based on the chemical and physical properties, minerals are broadly classified as Metallic and Non-metallic.
- Metallic minerals are the source of metals like iron, copper, gold, etc.


Minerals
Metallic Minerals
Non-metallic Minerals
Iron
Non-Iron
Ferrous Minerals
(Iron Ore, Manganese)
Non-ferrous Minerals
(Copper, Bauxite)
Organic or Energy Minerals
(Coal, Petroleum)
Inorganic Minerals
(Mica, Graphite, Limestone)

Mineral Resources

- Metallic minerals are further categorised as **ferrous and non-ferrous minerals.**
- **Ferrous mineral are those which have iron content like iron ore whereas non-ferrous minerals do not have iron content like copper, bauxite, etc.**
- **Non-metallic minerals are either organic or inorganic in origin.**

Category	Sub-type	Meaning	Examples
Metallic	Ferrous	Minerals containing iron content	Iron ore, Manganese, Chromite
Metallic	Non-Ferrous	Minerals with NO iron content	Copper, Bauxite, Gold, Lead, Zinc
Non-Metallic (Organic)	Mineral fuels / Fossil fuels	Derived from buried plant & animal remains	Coal, Petroleum
Non-Metallic (Inorganic)	—	Non-metal, non-fuel minerals	Mica, Limestone, Graphite, Gypsum

Characteristics of Minerals

- Uneven distribution — minerals are not spread equally over the earth's surface; they occur in patches/belts.
- Inverse quality-quantity relationship — good quality (high-grade) minerals are found in smaller quantities; poor quality (low-grade) minerals are found in larger quantities.
- Exhaustible — minerals take millions of years to form geologically and cannot be replenished quickly once used up. Hence they have "no second crop" and must be conserved.

Characteristics of Minerals

- **Impurities**— most minerals occur mixed with other substances and need processing before use.
- **No country is self-sufficient**— every country depends on others for some minerals (e.g., India imports copper).

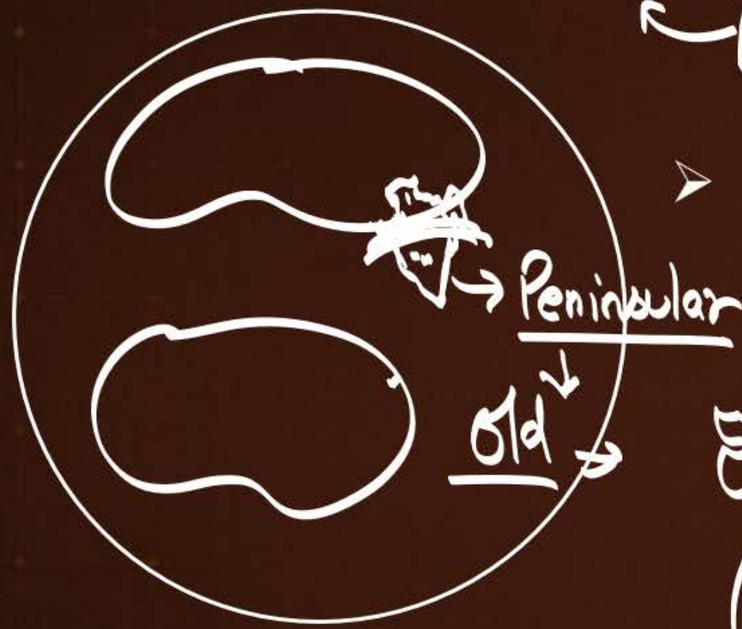
Why are Minerals Important to India's Economy?

- Economic development of a country is directly linked to mineral availability.
- Minerals form the raw-material base for large-scale industries (iron & steel, aluminium, cement, etc.).
- Agriculture also depends on minerals — used to make chemical fertilisers.

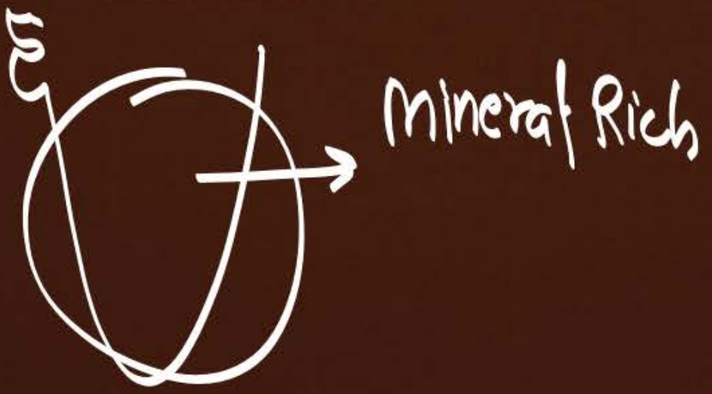
Why are Minerals Important to India's Economy?

Rock structure

- India has a rich variety of minerals because of its varied geological structure (old Peninsular rocks are mineral-rich).



- India ranks high among world producers of iron ore, coal, mica and manganese, and has large reserves of chromite, bauxite and limestone.



Iron ore

Iron Ore



- Iron ore is one of the most widely distributed minerals found in the Earth's crust. It forms the backbone of modern civilisation and the foundation of our basic industry.

- In India, the main varieties of iron ore found are:

Type	Colour	Iron Content	Special Feature	Where Found in India
Haematite ("Red Ore")	Reddish	60% – 70%	Most widely used industrial ore in India	Odisha, Jharkhand, Chhattisgarh, Karnataka, Maharashtra, Goa, Andhra Pradesh, Rajasthan
Magnetite ("Black Ore") → Finest	Dark brown to black	More than 70% (finest quality)	Has magnetic property — hence the name	Tamil Nadu, Karnataka (Kudremukh)
Limonite	Yellow / light brown	35% – 50%	Easiest & cheapest to mine (soft ore)	Raniganj coalfield, Garhwal (Uttarakhand), Mirzapur (UP), Kangra Valley (HP)
Siderite	Grey/brown	Less than 40%	Lowest grade, has impurities	Minor occurrences

Remember iron-content order highest→lowest: MAGNETITE > HAEMATITE > LIMONITE > SIDERITE = "My Home Likes Sweets" (>70%, 60-70%, 35-50%, <40%).

Marvel Hero Loki Superb



Magnetite (Fe_3O_4)



Hematite (Fe_2O_3)



Limonite ($2Fe_2O_3 \cdot 3H_2O$)



Siderite ($FeCO_3$)

Iron Ore

(1) **Magnetite:** It is dark brown to blackish in colour and is, therefore, known as 'Black Ore'. It is the best quality of iron and contains more than 70per cent of iron. Since it possesses magnetic property, it is called magnetite. It is found in Tamil Nadu and Karnataka.

Iron Ore

largest used

(2) **Haematite:** It is reddish in colour and is also known as 'Red Ore'. It contains 60 to 70per cent pure iron and is found in Odisha, Jharkhand, Chhattisgarh, Karnataka, Maharashtra, Goa, Andhra Pradesh and Rajasthan.

Iron Ore

(3) **Limonite:** It is yellow or light brown in colour and contains 35 to 50per cent of iron. It occurs in the iron stone group in the Raniganj coalfield. It is also found in Garhwal (Uttarakhand) and Mirzapur district (UP) and the Kangra valley (Himachal Pradesh).

Uses of Iron Ore

Alloy → Mixing of two metal

1. About 90% of iron ore mined is used to make STEEL (alloyed with other elements for strength & hardness) — used in construction, automobiles, trains & tracks.

2. Powdered iron → metallurgy products, magnets, electrical equipments , auto parts.

- Radioactive iron → used in medicine as a tracer element, and in biochemical/metallurgical research.

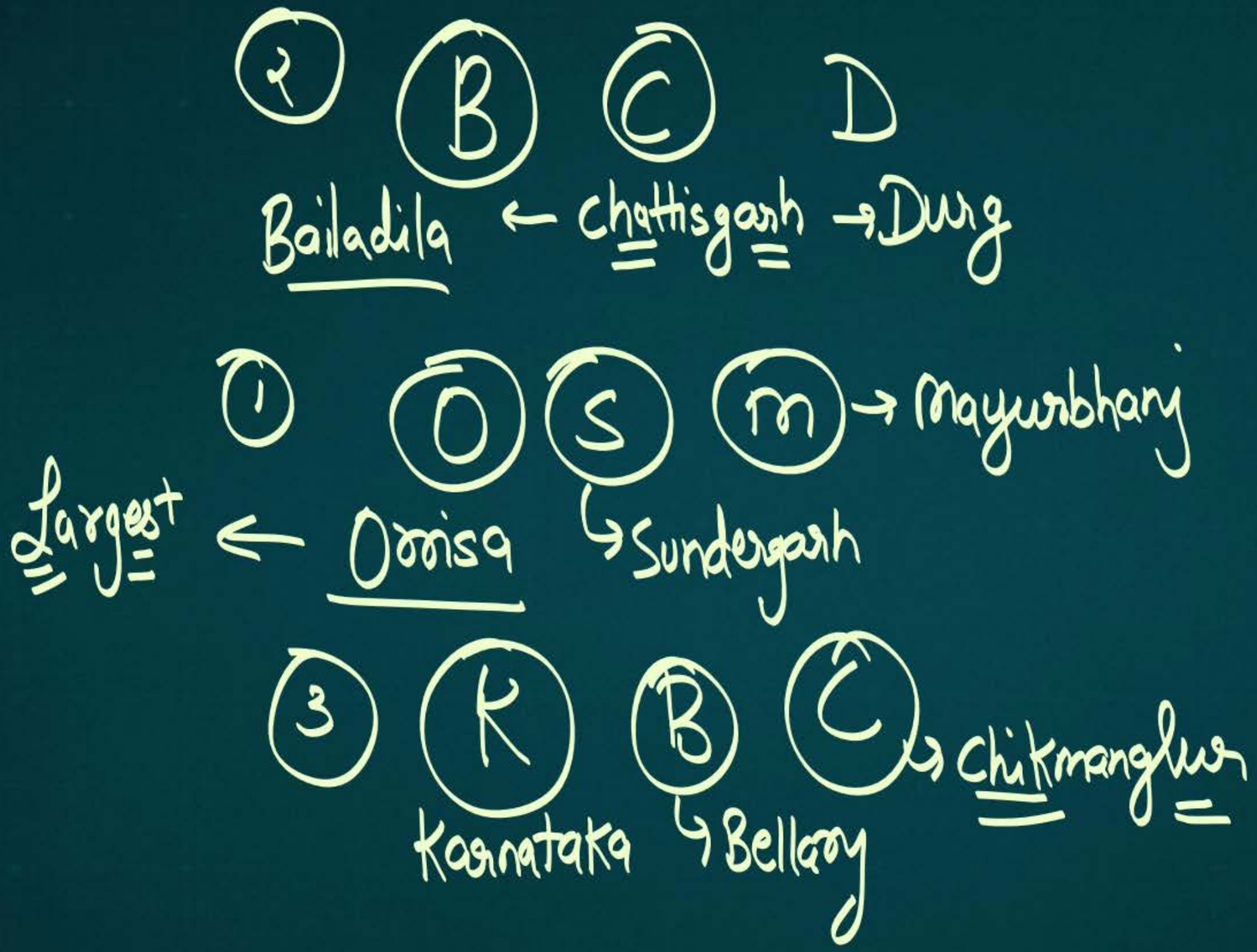
Uses of Iron Ore

3

- Iron blue (a pigment) → paints, printing inks, plastics, cosmetics (eye-shadow), artists' colours, enamel finishes.
  ↳ Paints
- Black iron oxide → pigment in polishing compounds, metallurgy, medicine, magnetic inks.

Distribution

State	Important Belts / Districts	Notes
Chhattisgarh	Bailadila (Dantewara), Dalli-Rajhara (Durg)	Supplies the Bhilai steel plant
Jharkhand	Chiria, Noamundi, Gua (all in Singhbhum district); Palamau	Chiria has one of the largest deposits
Odisha	Keonjhar, Mayurbhanj, Sundergarh, Sambalpur, Cuttack, Koraput	Haematite variety, 55–68% iron; feeds Rourkela, Bokaro, Durgapur, Jamshedpur, Asansol steel plants
Karnataka	Bababudan Hills (Chikmagalur), Bellary, Hospet, Shimoga, Chitradurga; Kudremukh (1000 million tonnes)	Rich in magnetite + haematite; feeds Bhadravati Iron Works
Goa	North Goa (richer deposits)	Inferior limonite & siderite variety; exported
Andhra Pradesh	Anantapur, Krishna, Kurnool, Cuddapah, Nellore	55–62% iron content
Tamil Nadu	Salem, North Arcot, Tiruchirapalli, Coimbatore, Madurai	—
Maharashtra	Ratnagiri, Chandrapur, Bhandara	—
Rajasthan	Moriza (Bhilwara), Udaipur	—


2
B
C
D
Bailadila ← Chattisgarh → Durg
1
O
S
m → Mayurbhanj
Largest ← Orrisa
↳ Sundergarh
3
K
B
C
↳ Chikmanglur
Karnataka
↳ Bellary

Distribution

Belt	States Covered	Key Areas
Odisha–Jharkhand Belt	Odisha, Jharkhand	Badampahar, Mayurbhanj, Keonjhar, Singhbhum (Noamundi, Gua)
Durg–Bastar–Chandrapur Belt	Chhattisgarh, Maharashtra	Bailadila — super high-grade haematite exported via Vishakhapatnam
Ballari–Chitradurga–Chikkamagaluru–Tumakuru Belt	Karnataka	Kudremukh, Bellary, Hospet
Maharashtra–Goa Belt	Maharashtra, Goa	Ratnagiri, North Goa — lower grade, but easily exported via sea

- **Odisha is currently INDIA'S No.1 iron-ore producing state (over 50% of national output), followed by Chhattisgarh and Karnataka.**


India: Minerals - Iron Ore
Iron Ore
Bhilwara
Udaipur
Tropic of Cancer
Palamau
INDIA
Singhbhum
Mayurbhanj
Sundergarh
Keonjhar
Durg
Sambalpur
Cuttack
Chandrapur
Daitewada
Koraput
ARABIAN SEA
BAY OF BENGAL
Ratnagiri
North Goa
Bellary
Krishna
Anantapur
Nellore
Shimoga
Chitradurga
Cuddapah
Chikmagalur
Salem
Tiruchirapalli
Coimbatore
Madurai
LAKSHADWEEP (INDIA)
LAKSHADWEEP SEA
ANDAMAN AND NICOBAR ISLANDS (INDIA)
ANDAMAN SEA
INDIAN OCEAN
SCALE 1:24 000 000
Km 0
240
480
720 Km


Manganese

Manganese

- It is a tough, dark-colored metal that looks very similar to iron and is found naturally in the earth combined with oxygen (as a natural oxide).
- It plays a key role in smelting (the heating and melting process used to extract pure metal from raw iron ore).
- It is used to create ferro alloys, which are special mixtures of iron and other metals combined to make steel stronger, harder, and more durable.

Uses of Manganese

- Makes steel tough & rust-resistant — its most important use (iron & steel industry).
- Used to manufacture black enamel.
- Used in chemical industry to manufacture bleaching powder, insecticides & paints.
- Used in electrical and glass industries.
- Used to form several important alloys (ferro-manganese).

Uses of Manganese

- Used in manufacturing dry-cell batteries.
- Essential for plant growth (helps reduce nitrates in green plants/algae) and for animal/human metabolism (enzyme action, bone development) in small quantities — excess is toxic.
- Manganese can also regulate blood sugar levels and supports the immune system. It is also involved in bone development and reproduction.


India: Manganese
Manganese
Pali
Udaipur
Tropic of Cancer
Banswara
Palamau
Jabalpur
Mandla
Balaghat
Singhbhum
Chaibasa
Mayurbhanj
Sundergarh
Keonjhar
Sambalpur
Bonai
Talcher
Kalahandi
Koraput
Srikakulam
Vijayanagaram
Adilabad
Sanguem
North Kanara
Bellary
Davanagere
Shimoga
Chitradurga
ARABIAN SEA
BAY OF BENGAL
LAKSHADWEEP SEA
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ANDAMAN AND NICOBAR ISLANDS (INDIA)
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720 Km

Distribution of Manganese in India

Distribution

Manganese ore found in India is of good quality and therefore, is in great demand in the world market. USA, Japan, UK, France, the Netherlands, West Germany and Belgium are its main buyers.

Manganese

Andhra Pradesh	Srikakulam, Vijayanagaram.
Telangana	Adilabad.
Goa	Sanguem.
Jharkhand	Singhbhum, Palamau, Chaibasa.
Karnataka	Sandur, Shimoga, Bellary, Chitradurga, Tumkur, Belagavi, North Kanara, Davangere.
Madhya Pradesh	Chhindwara, Balaghat, Mandla, Jabalpur.
Maharashtra	Nagpur, Bhandara.
Odisha	Keonjhar, Mayurbhanj, Talcher, Sundargarh, Bonai, Koraput.
Rajasthan	Banswara, Udaipur, Pali.

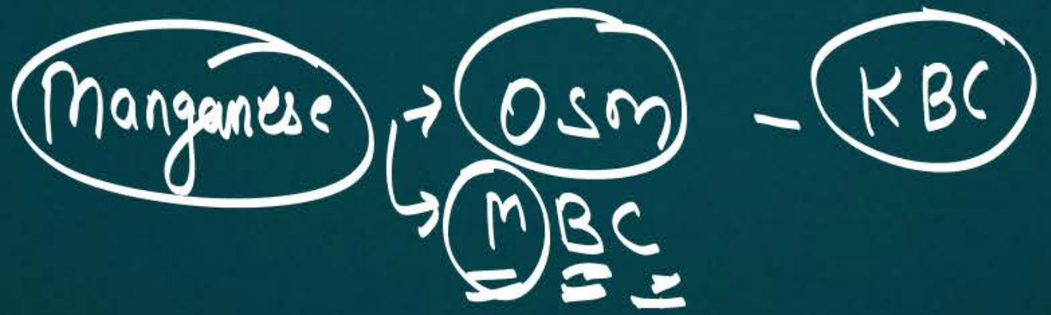
OSM MCB

State	Important Mining Areas
Odisha (LARGEST producer) -1	Keonjhar, Mayurbhanj, Talcher, Sundargarh, Bonai, Koraput
Madhya Pradesh - 2	Chhindwara, Balaghat, Mandla, Jabalpur
Maharashtra - 3	Nagpur, Bhandara
Karnataka	Sandur, Shimoga, Bellary, Chitradurga, Tumkur, Belagavi, North Kanara, Davangere
Andhra Pradesh	Srikakulam, Vijayanagaram
Telangana	Adilabad
Goa	Sanguem
Jharkhand	Singhbhum, Palamau, Chaibasa
Rajasthan	Banswara, Udaipur, Pali

Odisha = single most important manganese state (about 1/3rd of national output).

Joon ore → Marvel Hero Loki Superb

- O S M
- B C D
- K B C




Saturated

Thank You